

Aegis-CNI

AI cyber resilience for critical national examination infrastructure.

Detect	Predict	Attribute	Simulate	Mitigate	Audit
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Protected asset: CBSE-style digital examination portal with candidate, invigilator, examiner, certificate, and SOC workflows.

Submission claim: live attacks are converted into scored telemetry, ATT&CK attribution, source blocking, digital-twin simulation, and hash-chained audit evidence.

Why This Matters

Exam infrastructure is a public-trust system. Breach or downtime can disrupt students, institutions, results, and national confidence.

- Signature-only security reacts after known compromise patterns exist.
- Low-and-slow APT behaviour requires behavioural intelligence and sequence analysis.
- Zero-day posture must rely on anomaly, prediction, containment, and auditability.
- Judges can verify the system live using terminal attack commands, not just seeded screenshots.

What We Built

A full cyber-resilience loop around a running portal and FastAPI backend.

Frontend CBSE-style portal, MFA login, SOC dashboard, digital twin, Copilot	Backend FastAPI middleware, ML scoring, ATT&CK RAG, SOAR, audit
GenAI Online Ollama, local Ollama fallback, offline deterministic fallback	Controls IP block, redirect guard, spoofing guard, password policy, AI guard

AI/ML Pipeline

The alert decision is based on behaviour, not only database lookup.

- IsolationForest anomaly scoring over telemetry features.
- Z-score baseline deviation for latency, bytes, role, segment, and event pattern.
- Predictive risk model using rare transitions and short-window entity pressure.
- MITRE ATT&CK retrieval over a local Enterprise ATT&CK corpus.
- GenAI attribution produces defensive evidence, likely next stage, and mitigation guidance.

Live Mitigation Coverage

Aegis-CNI actively blocks selected live web, identity, redirect, spoofing, and AI-prompt attack classes.

SQLi / command injection	XSS / path traversal	Brute force / scan burst
Open redirect	IP spoofing / untrusted XFF	AI prompt injection

Detected live attacks become telemetry, alerts, ATT&CK mappings, active source blocks, and audit records. Follow-up requests return HTTP 403.

GenAI Safety

The system protects the AI surface as well as the web surface.

- SOC Copilot answers only portal-security and incoming-attack questions.
- Prompt injection, jailbreak, system-prompt leakage, tool override, and API-key exfiltration prompts are blocked before GenAI execution.
- Provider chain: online Ollama -> local Ollama -> offline deterministic fallback.
- Output is defensive-only: evidence, likely next stage, containment, verification, and recovery.

Cyber Resilience Digital Twin

Graph-based what-if simulation for attack paths and impact analysis.

- Models assets: perimeter/WAF, candidate portal, certificate service, auth DB, marking portal, proctoring, SOC.
- Tracks asset criticality, controls, live alert pressure, and blocked source count.
- Simulates likely attack path and risk-before/risk-after containment posture.
- Model-only by design: no live user session or production network is touched.

Demo Flow

A five-minute judge demo proves the platform is live.

- Sign in as SOC operator with MFA.
- Run a safe terminal attack, such as SQLi or AI prompt injection.
- Watch live alert, score, latency, bytes, attribution, and blocklist update.
- Show follow-up request returns HTTP 403.
- Ask SOC Copilot: 'Why was this blocked?'
- Run digital twin simulation and verify `/audit/verify``.

Measured Impact

Generated by the repository evaluation harness.

Detection Rate	ATT&CK Accuracy	MTTD Improvement	MTTR Improvement
80.56%	100.00%	10080.0x	68.6x
Local ATT&CK techniques	Labelled attacks	Autonomous playbook steps	Chroma ready
697	108	75.0%	True

Production Readiness

Honest path from hackathon prototype to critical-infrastructure deployment.

- Already implemented: live middleware, ML scoring, GenAI attribution, source blocking, MFA demo, password policy, digital twin, audit verification.
- Enterprise integrations needed: SIEM, EDR, IAM, WAF/firewall, cloud audit, CMDB, notification systems, Redis blocklist, observability.
- Submission includes deployment configs, demo scripts, judge Q&A, one-page summary, and a generated 90-second pitch video.

Submission Links

Ready for ET GenAI Hackathon review.

GitHub: https://github.com/AnandSaurabhShah/ET_HACK2

Demo credentials: SOC-AEGIS-001 / security

Key docs: README.md, DEPLOYMENT.md, PRODUCTION_READINESS.md, RESULTS.md, submission/

Closing line: Aegis-CNI demonstrates the resilience loop ET GenAI asks for: detect, predict, attribute, simulate, mitigate, and prove.